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Fundamentals of Programming and Computer Science

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TP3 Report (Final)

Project:

Labyrinth (formerly “Mines of Ibdineth”)

Description:

This game follows an adventurer who journeys into an ever changing maze. The adventurer is represented by a sprite, which has the ability to mine blocks and open doors (given that it has a key). By mining blocks, the adventurer collects currency, which can be redeemed at the store to buy a key that allows the player to proceed to the next area. A player’s progress is marked by how close it is to the treasure chest, which is indicated by the types of keys the player has.

The ultimate goal of the player is to get to the end of the map and reach the treasure, which will give the player the time they took to get to the treasure. This time can be used as a progress check for the future times they play the game so that they can see how their speed improves.

Minecraft:

-Random Environment Generation: Minecraft, by default, spawns the player in a random biome.

In my game, the player will always spawn at the mine entrance, but the map shifts every few seconds, making it hard to navigate.

-Resource System: Like in Minecraft, my game will keep track of the number of each resource, changing based on if I have obtained more of that material or if I have used it.

Mining FRVR:

-Character animations: In Mining FRVR, the character moves dynamically when going in certain directions. I would like to implement these into my code more to make the mining and moving experience more authentic.

-Store/Currency System: Like in Mining FRVR, my game will involve gaining currency through exploring. This currency will then be used to buy objects at the store (another feature also in Mining FRVR), which is represented by a house in the main game page.

-Equipment Type/Category: Mining FRVR splits equipment into two categories, clothing and tools. Similarly, my game will include keys as equipment my adventurer can equip and use.

-Interaction with objects: In Mining FRVR, the player can push objects to cause certain events to happen (connecting 3 blocks together causes a similar effect to TNT). In my game, I would like to be able mine and open doors.

Terraria:

-Leveling up/game progression: In Terraria, the player levels up by fighting enemies. In my code, I would like to modify that by adding that I can gain currency by mining blocks. After gaining enough currency, the player can buy a key that can be used to progress to the next area.

-Player guide system: Terraria has an NPC (non-playable character) that guides the player through certain mechanics. I would also like to add a similar kind of player guide for each new mechanic encountered to ensure that the player understands what to do, like a help page players can refer to.

Version Control/Backup Plan

I will copy a version of my code to Google Drive, listing it by date and version number in order to keep track of it:

Tech Demo:

I will not be using additional modules or external hardware to complete my project.